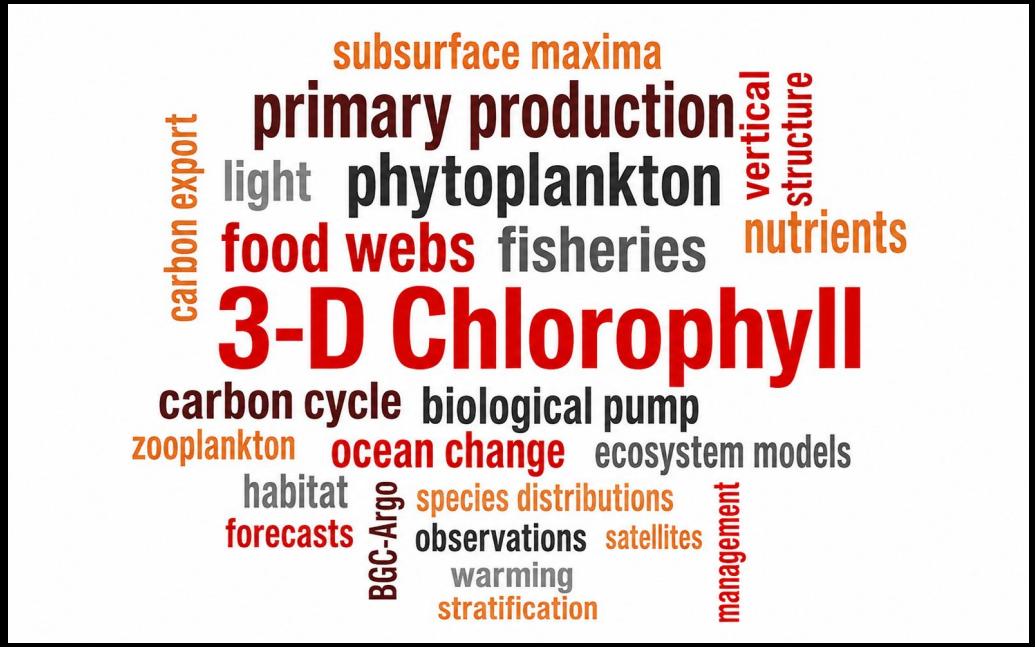
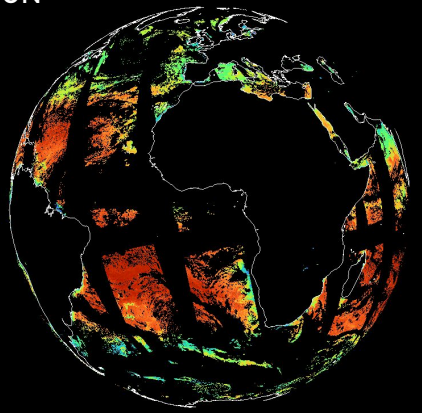


GO-BGC HACKWEEK 2026

ELI HOLMES
JODI BREWSTER
CHARLIE TURNER
ANDREA ALBRIGHT
PETER MASTNAK
HAIPENG ZHAO
NABIR MAMNUN
KAI YANG



3-D CHLA from ML + ARGO

BACKGROUND, the basic idea for 3D CHLA

01 Data Integration

In-Situ Matchups

In-situ CHLA at 10m depth bins to 200m matched with PACE 128 bands Remote-sensing reflectance (~5,500 profiles).

02 Model Training

Boosted Regression Trees

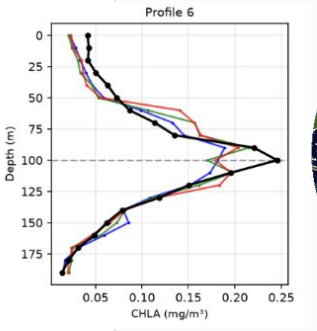
Trained on standardized CHLA at 10m depths to reconstruct vertical structure.

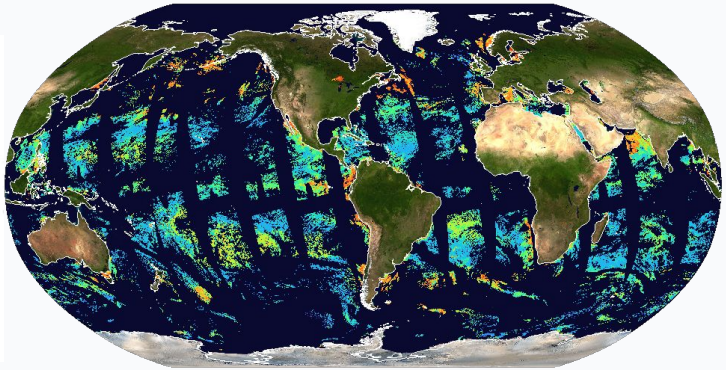
03 Application

Global Predictions

Generating high-resolution global 3D predictions directly from daily PACE mission data.

Mapping global CHLA at 10m depths, the depth of peak CHLA, and peak height from **March 2024 to present**.

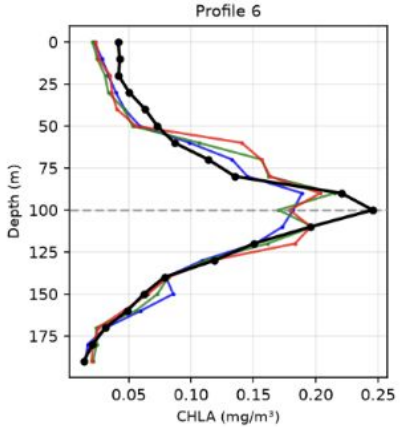




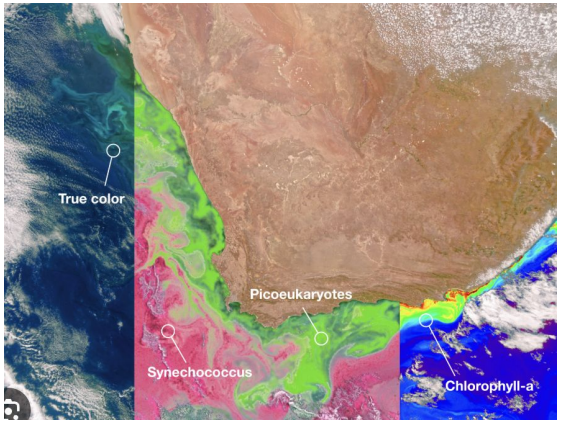
Ultimate goal: better 3D CHLA estimates. Daily, closer to coast, better accuracy, open model.



Does hyperspectral help? SOCA2020 (Copernicus) product used multispectral bands.
Does training on CHLA help?



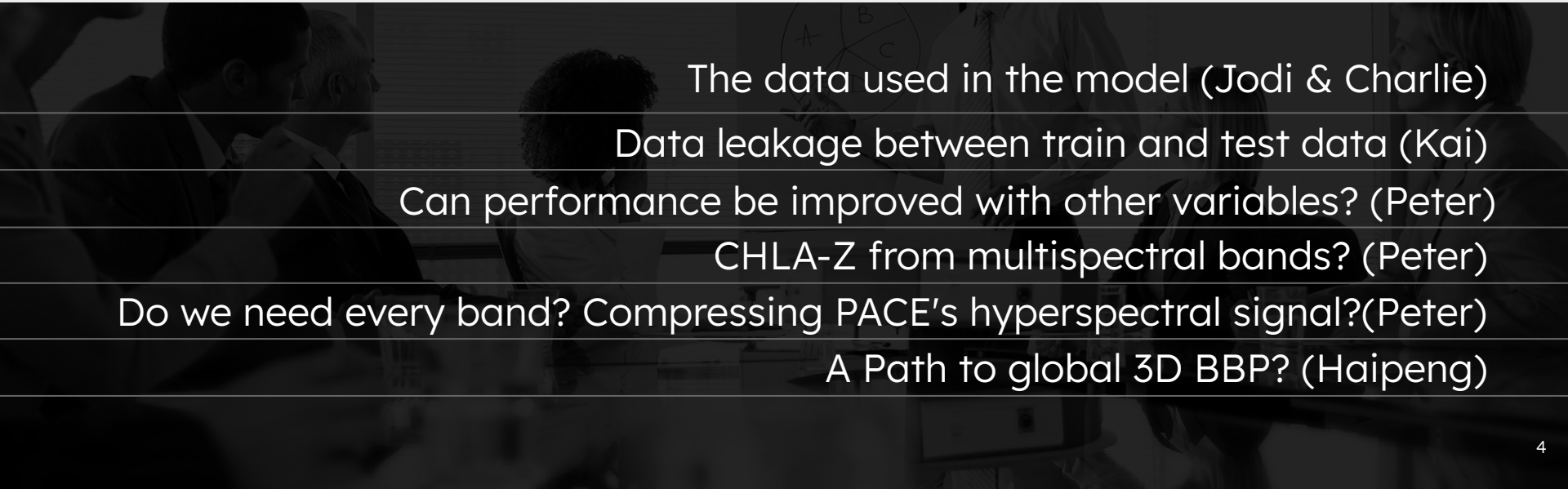
Does ocean color contain information about vertical chlorophyll structure beyond that already contained in surface chlorophyll?



Can we resolve closer to the coast?

Existing products and methods are for open ocean.

WHAT WE EXPLORED



The data used in the model (Jodi & Charlie)

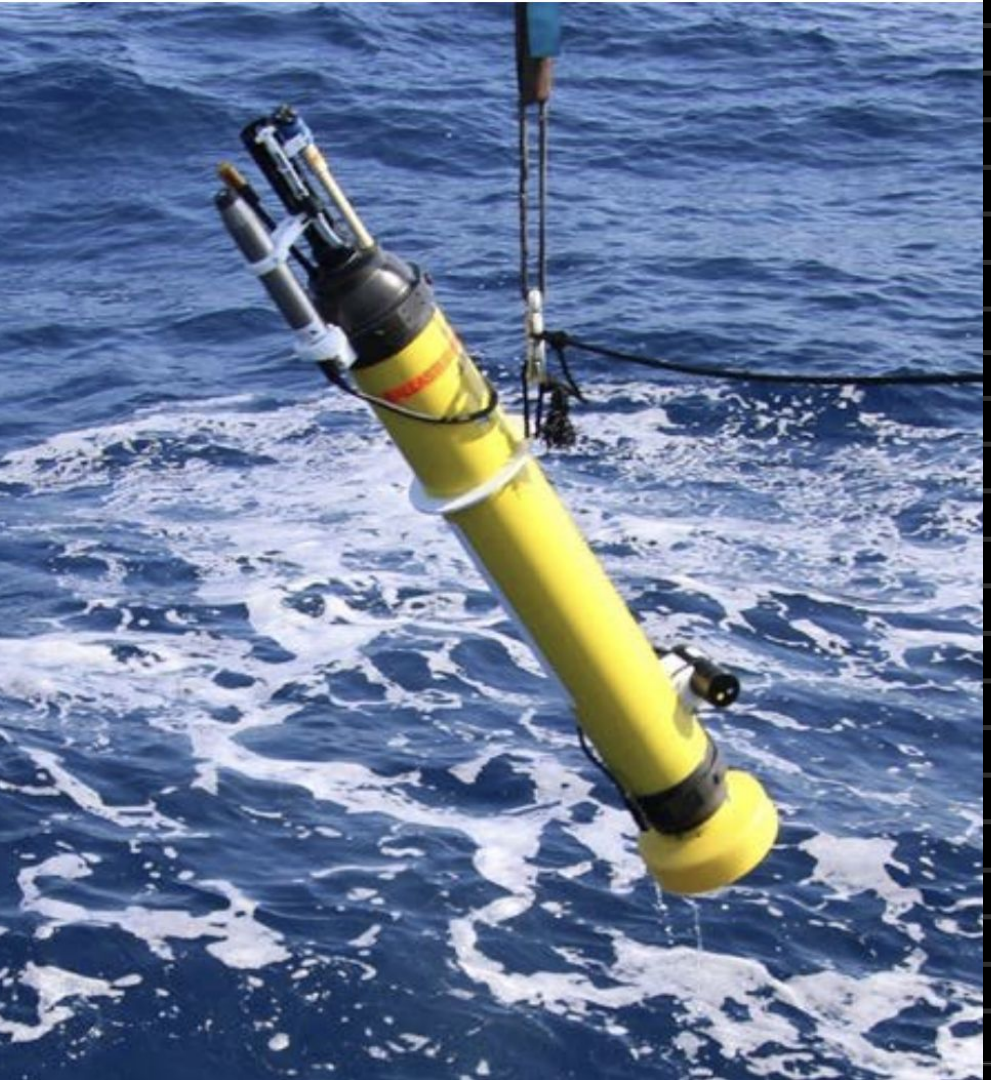
Data leakage between train and test data (Kai)

Can performance be improved with other variables? (Peter)

CHLA-Z from multispectral bands? (Peter)

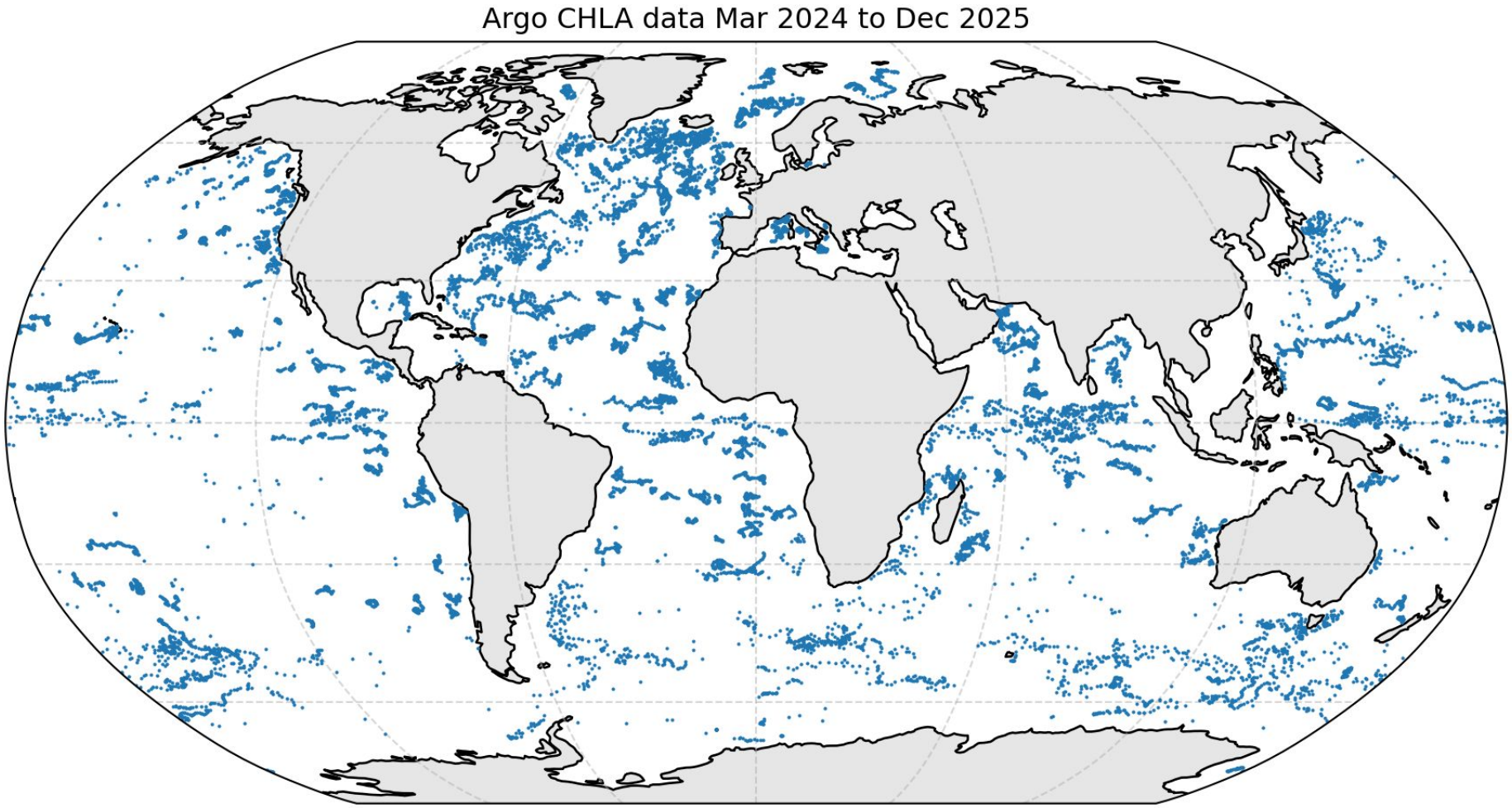
Do we need every band? Compressing PACE's hyperspectral signal?(Peter)

A Path to global 3D BBP? (Haipeng)



Argo

PRES	Pressure aka depth
TEMP	Temperature
CHLA	Chlorophyll from fluorescence
BBP	Backscattering (at different wavelengths)


Argo CHLA data Mar 2024 to Dec 2025



Ocean Observatories Initiative

Coastal Endurance Array

Coastal Pioneer Mid-Atlantic Bight Array

Global Irminger Sea Array

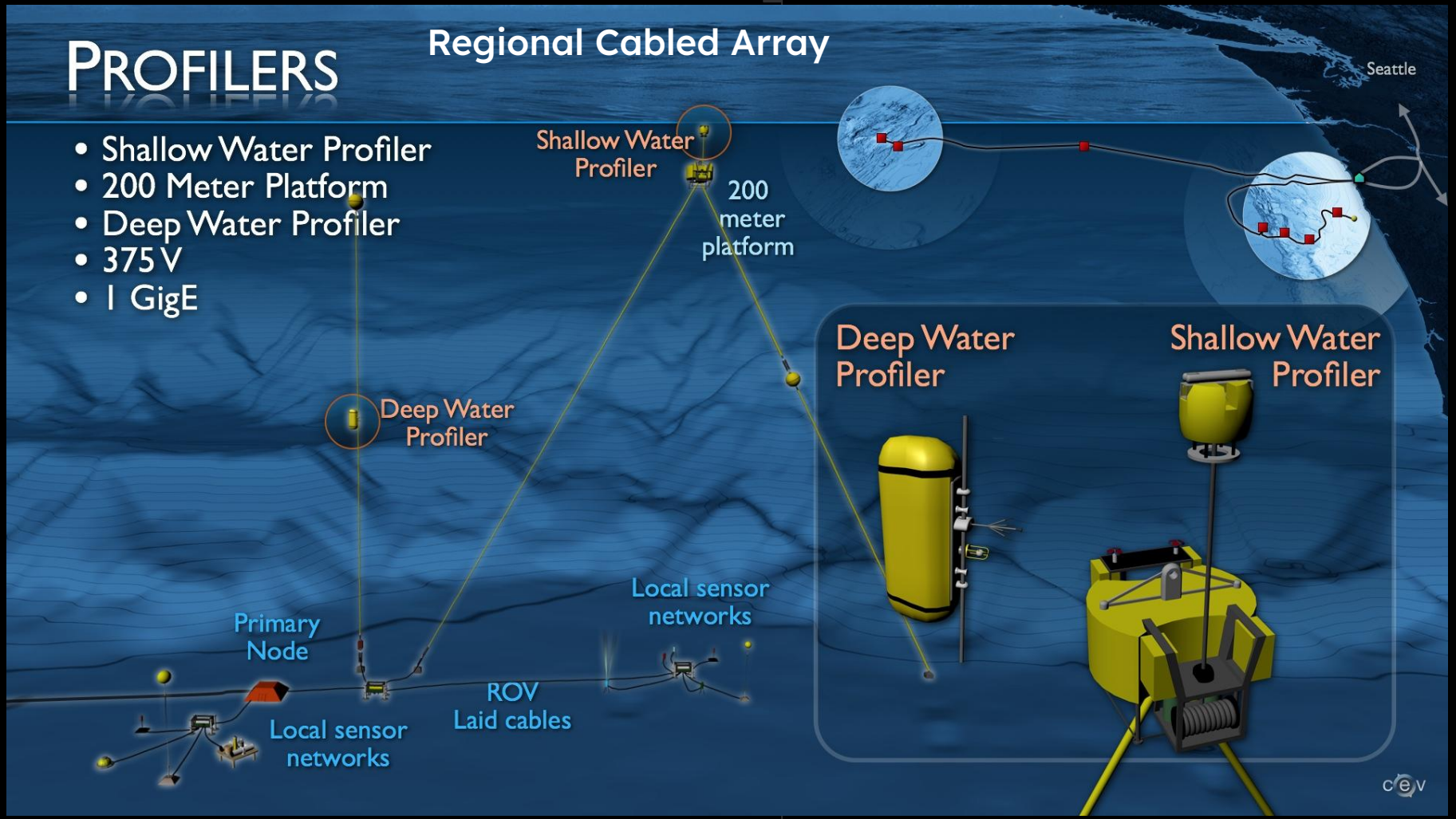
Global Station Papa Array

Regional Cabled Array

PROFILERS

Regional Cabled Array

- Shallow Water Profiler
- 200 Meter Platform
- Deep Water Profiler
- 375 V
- 1 GigE

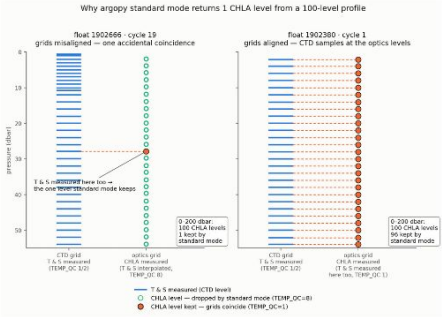


Before you train on CHLA-Z: what is actually in the training set

GO-BGC workshop, 2026-08-17 → 21 · drift experiments 26 (Argo half), 27 (OOI half), 28 (Argo ⇄ OOI match-ups) · all numbers reproducible from the READMEs

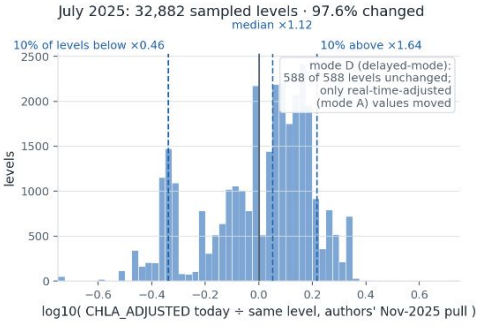
Charlie Turner
UC Davis · cjturner@ucdavis.edu

1 · The Argo half was 38% empty — and it's a default

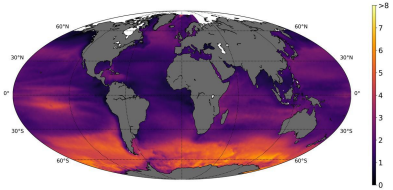


- argopy's default mode drops every CHLA level where T & S are interpolated (QC 8) — only grid coincidences survive. Still true today.
- Rebuilt from Sprof: coverage 62% → 98%. Fix: mode='expert', filter on CHLA_ADJUSTED_QC.

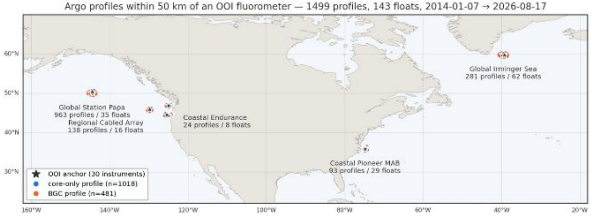
2 · CHLA_ADJUSTED moved under our feet — the 2026 Sauzède rollout



- 98% of values rewritten; the same level rescaled ×0.5 to ×1.6 by position. Mode D untouched.
- No history inside the files — a mode-A number carries a pull date. Pin a GDAC snapshot. OOI never got this correction.

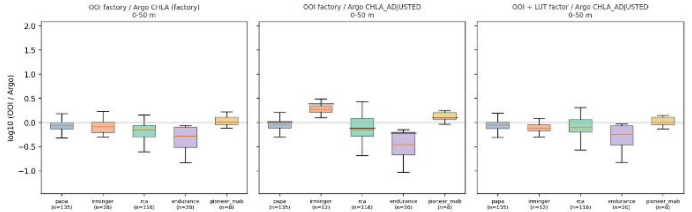


3 · Where Argo meets OOI — the match-up population



- 1,499 profiles / 143 floats within 50 km, 2014 → today (PACE era 249 / 23). Papa dominates.
- T/S agree to 0.05 °C / 0.01 PSU at Papa, Irminger, RCA — the pairing is sound. Pioneer MAB crosses a front, not a co-location.

4 · The sensors agree as factory fluorometers — not as one training target



- OOI factory ÷ Argo factory = 0.7–1.0×. After Argo's adjustment: 0.35× to 1.9× by site — the Sauzède factor (black bars).
- Divide OOI by the LUT → back to 0.76–1.00. Today's training target changes meaning by site; pick the basis explicitly.

PACE

Hyperspectral Rrs(λ)
Since 2024



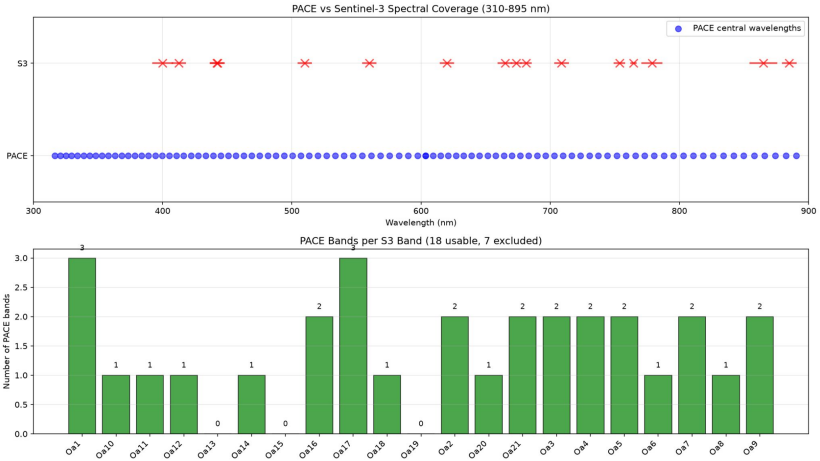
Sentinel 3

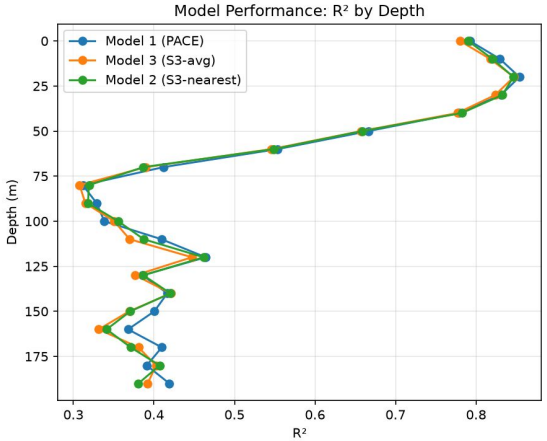
Multispectral Rrs(λ)
Since 2016

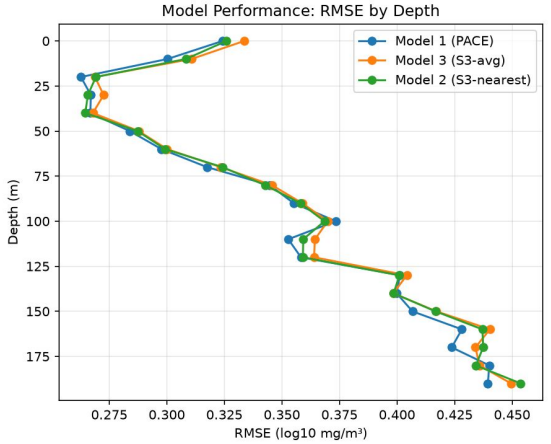


What happens when you resample PACE bands to match Sentinel-3 bands?

PACE data is recent (2024-now)
vs. S3 goes back to 2016

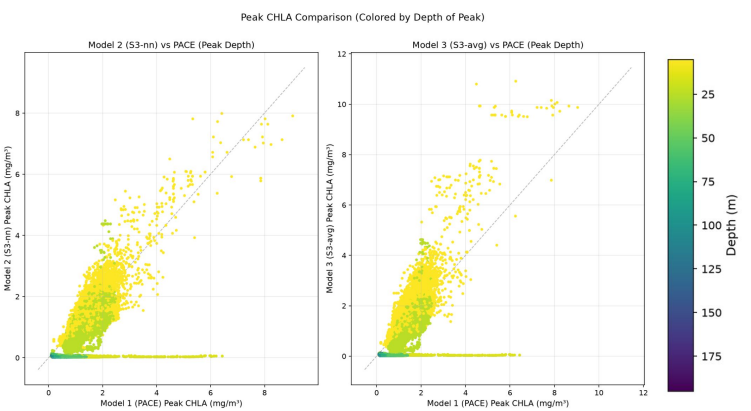


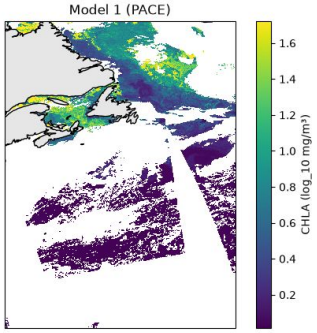


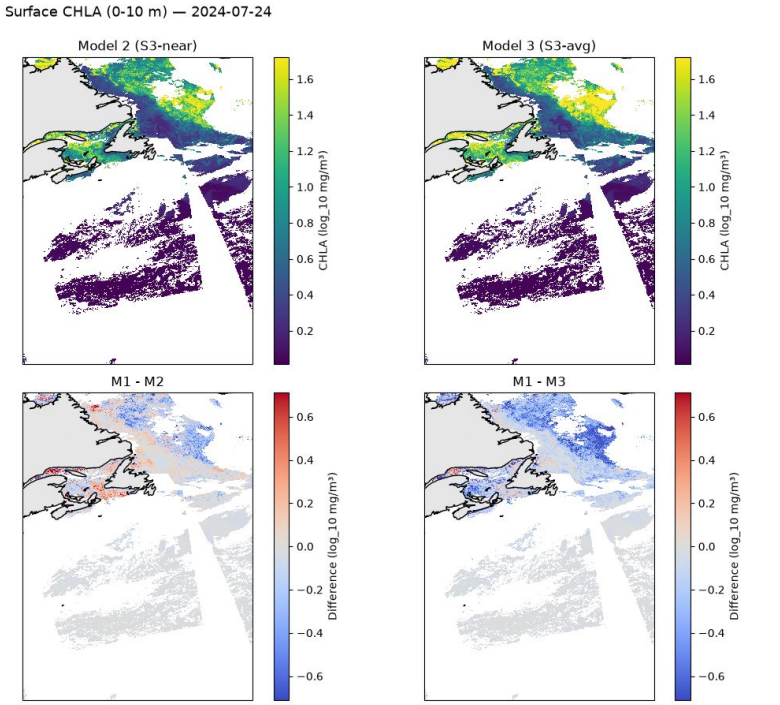


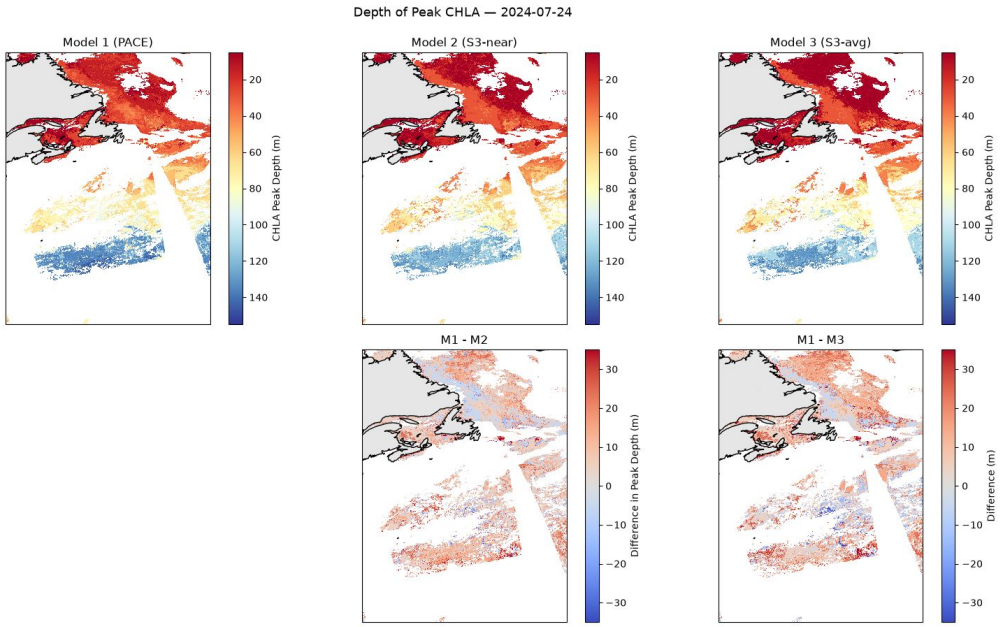


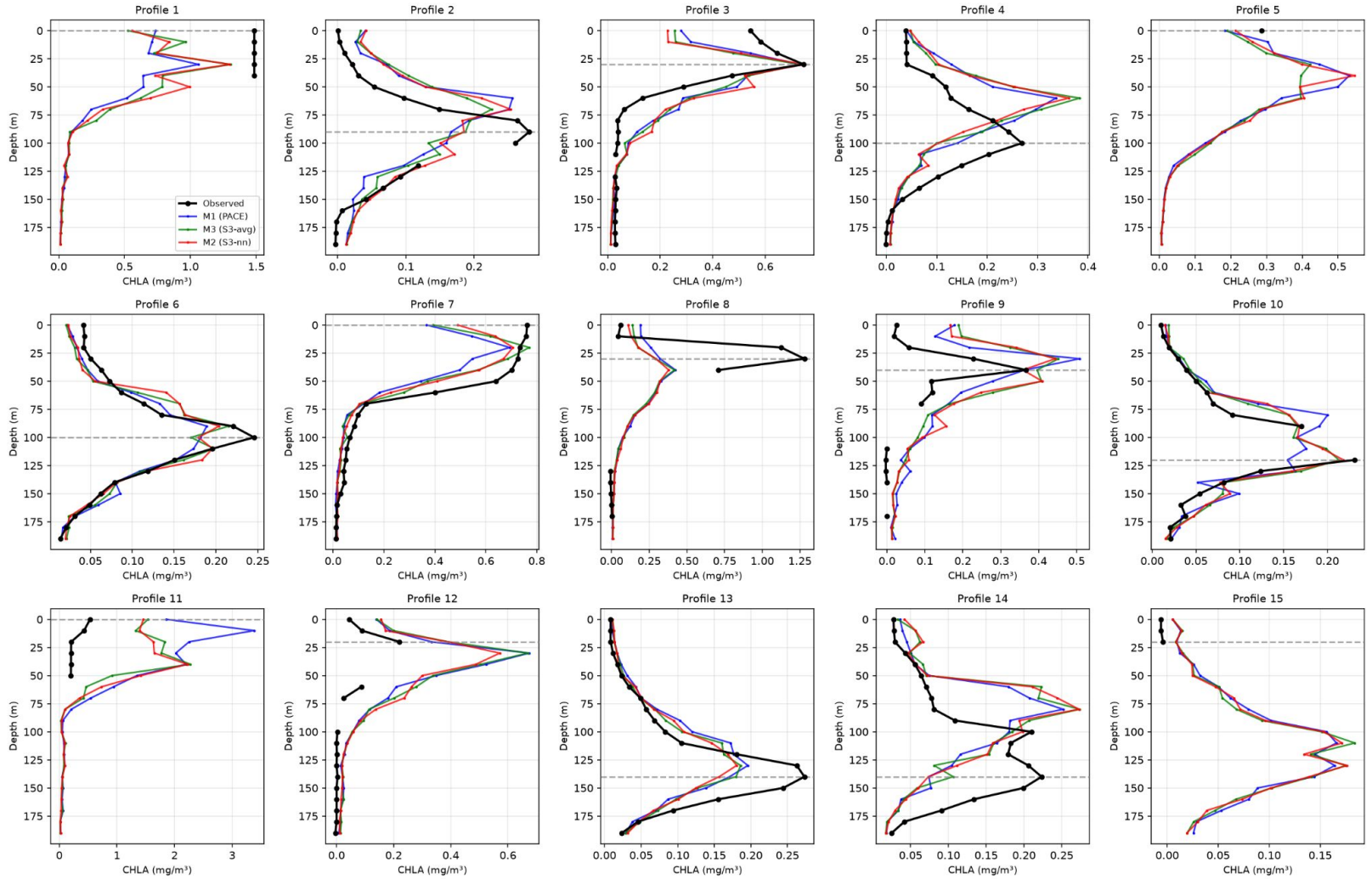



Peak CHLA Comparison (Colored by Depth of Peak)
Model 2 (S3-nn) vs PACE (Peak Depth)
Model 3 (S3-avg) vs PACE (Peak Depth)
Model 2 (S3-nn) Peak CHLA (mg/m³)
Model 3 (S3-avg) Peak CHLA (mg/m³)
Model 1 (PACE) Peak CHLA (mg/m³)
Depth (m)


Model 1 (PACE)
CHLA (log_10 mg/m³)

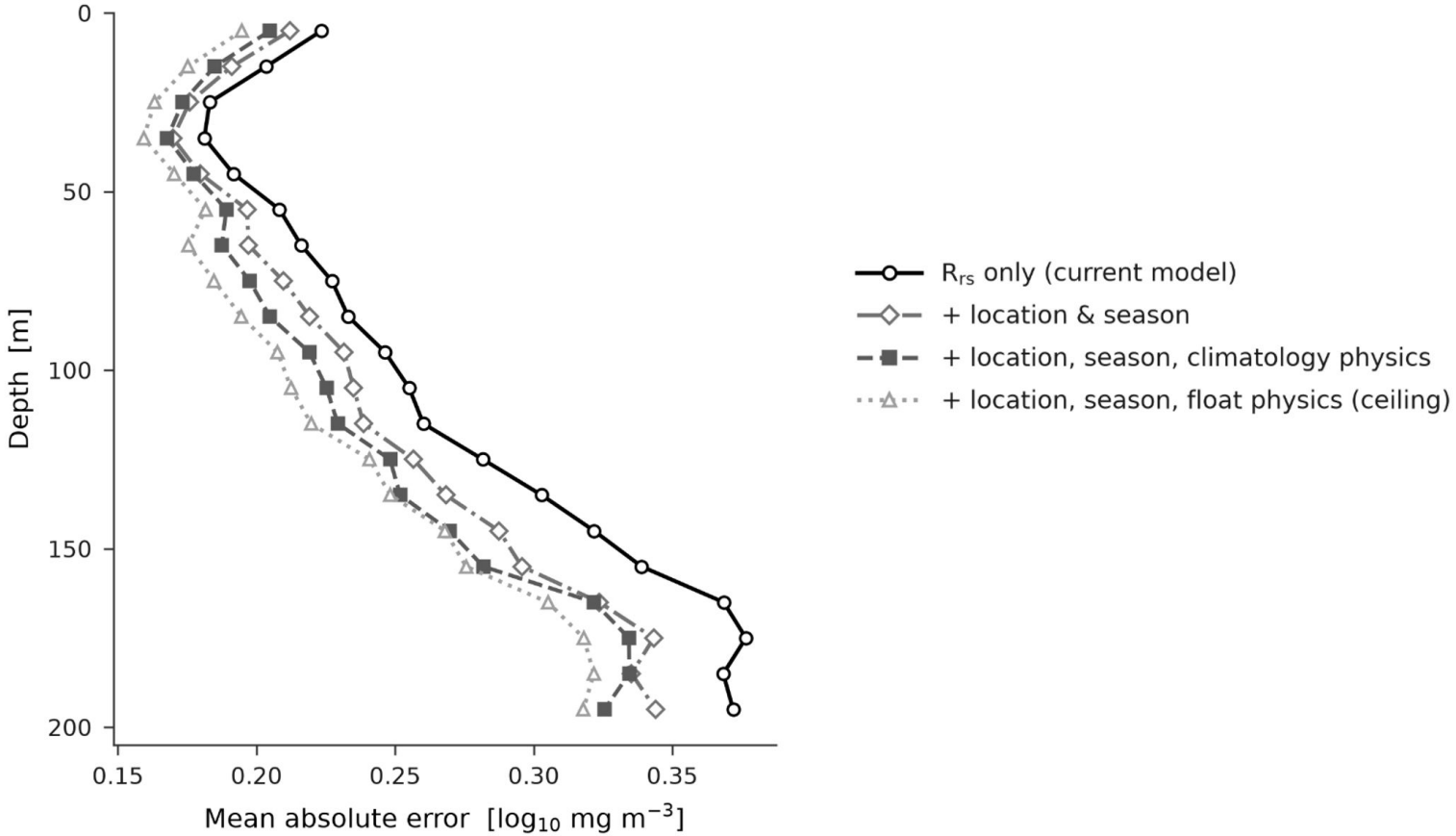

Surface CHLA (0-10 m) — 2024-07-24
Model 2 (S3-near)
Model 3 (S3-avg)
M1 - M2
M1 - M3
CHLA (log_10 mg/m³)
Difference (log_10 mg/m³)


Depth of Peak CHLA — 2024-07-24
Model 1 (PACE)
Model 2 (S3-near)
Model 3 (S3-avg)
M1 - M2
M1 - M3
CHLA Peak Depth (m)
Difference in Peak Depth (m)
Difference (m)

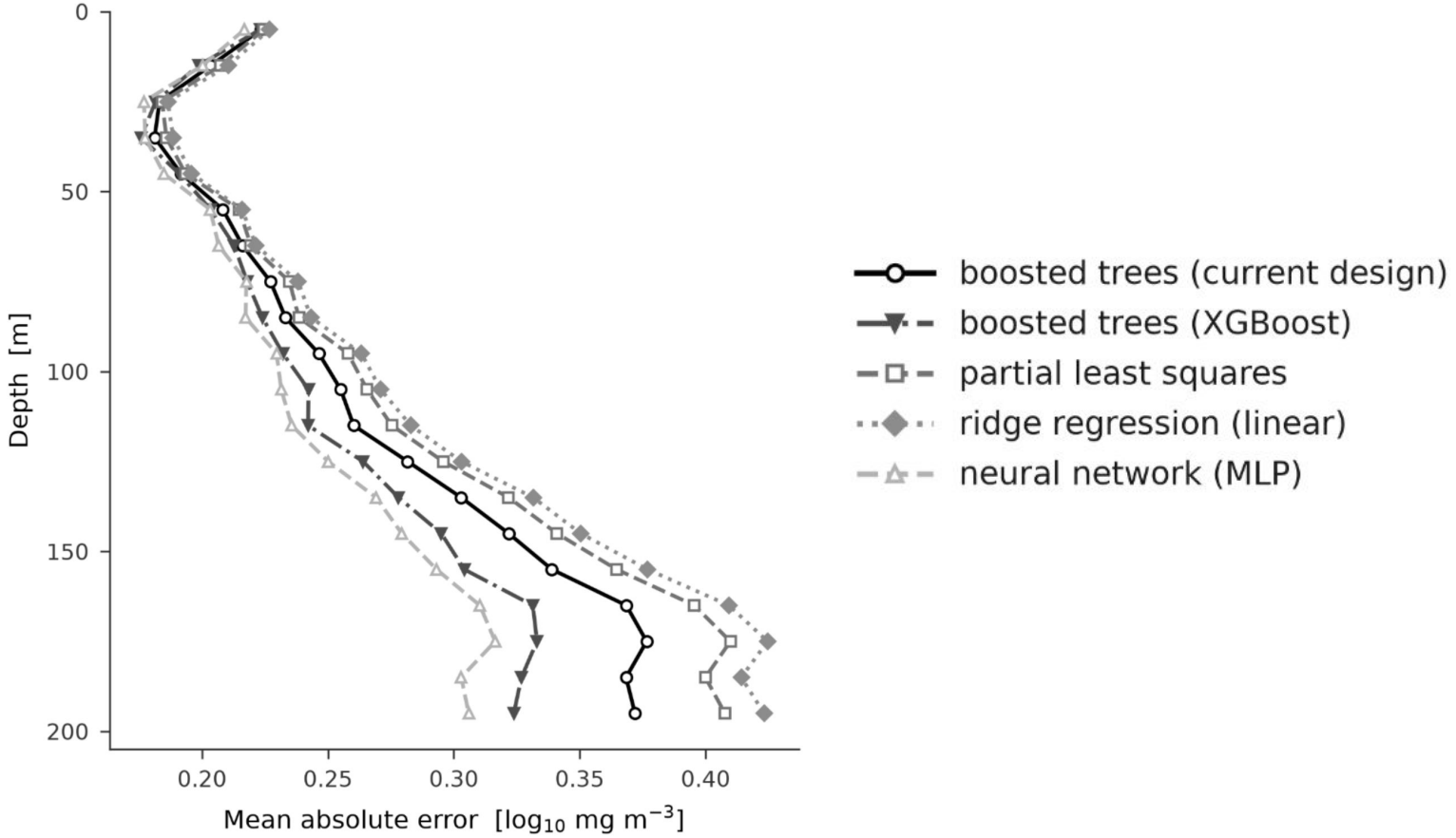

Random Argo Profiles with Model Predictions (Test Set)
Profile 1
Profile 2
Profile 3
Profile 4
Profile 5
Profile 6
Profile 7
Profile 8
Profile 9
Profile 10
Profile 11
Profile 12
Profile 13
Profile 14
Profile 15
Depth (m)
CHLA (mg/m³)
Observed
M1 (PACE)
M3 (S3-avg)
M2 (S3-nn)

Can performance be improved by adding other variables besides remote-sensing reflectance and how does performance change when testing different ML models?

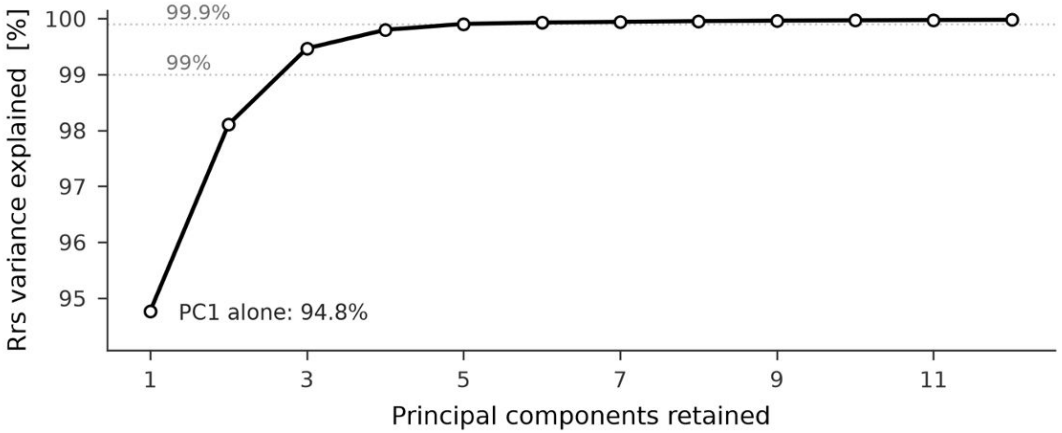
Example of how Tier A variables + location&season improve the model:



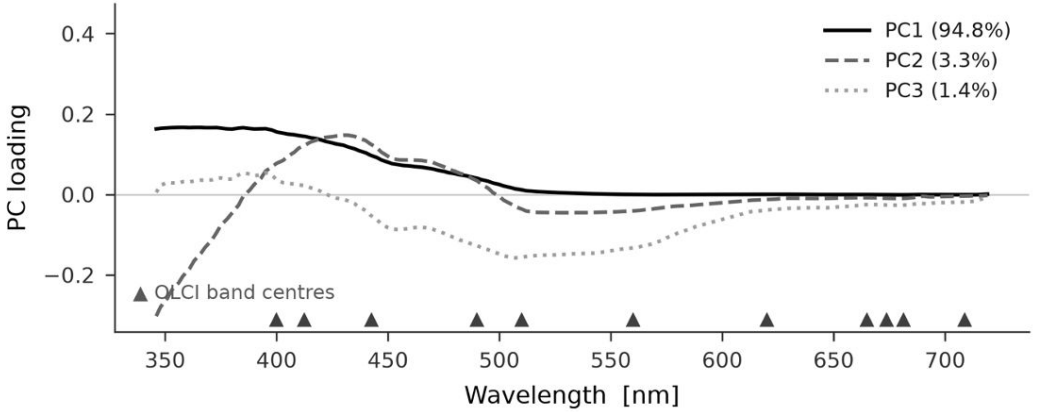
Testing performance of different ML models



Do we need every band? Three components capture 99.5% of reflectance



The spectral fingerprint of PC1-3



Adding physical and biogeochemical context improves skill

Variable	Description
Tier A — Physical — available globally from reanalysis or climatology	
`mld_dens`	Mixed-layer depth, density threshold 0.03 kg m^{-3} vs 10 dbar
`mld_temp`	Mixed-layer depth, temperature threshold 0.2 °C
`sst`	Conservative temperature at the shallowest level
`sss`	Absolute salinity at the shallowest level
`n2_max`	Maximum buoyancy frequency N^2 in the upper 300 m
`pycnocline_depth`	Depth of that N^2 maximum
`strat_index`	Potential energy anomaly over 0–200 m
`temp_200`	Conservative temperature at 200 m
`dens_diff_0_200`	Potential density difference, surface to 200 m
Tier B — Nutrient structure — globally available only as climatology	
`nitracline_depth`	First depth exceeding surface nitrate +1 μmol kg^{-1}
`nitrate_surf`	Mean nitrate above 20 dbar
Tier C — In-situ only — diagnostic ceiling, not deployable	
`doxy_surf`	Mean dissolved oxygen above 20 dbar
`doxy_min`	Minimum dissolved oxygen in the upper 1500 m
`doxy_min_depth`	Depth of the oxygen minimum
`aou_proxy`	Surface minus minimum oxygen (apparent utilisation proxy)
`bbp700_surf`	Particulate backscatter at 700 nm above 50 dbar

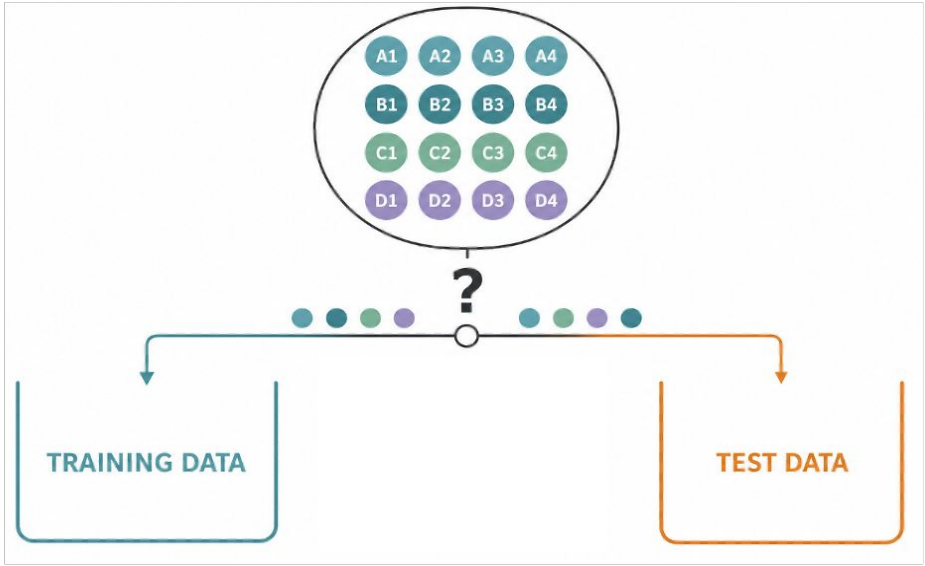
	Predictor set				
Depth (m)	R_{rs}	+A	+A+B	+A+B+C	Δ(+A)
0–10	0.775	0.815	0.820	0.823	**+0.040**
10–20	0.811	0.842	0.843	0.845	**+0.031**
20–30	0.840	0.863	0.866	0.863	**+0.023**
30–40	0.834	0.857	0.864	0.873	**+0.023**
40–50	0.777	0.811	0.831	0.842	**+0.033**
50–60	0.678	0.735	0.760	0.779	**+0.058**
60–70	0.544	0.665	0.688	0.687	**+0.121**
70–80	0.386	0.546	0.571	0.592	**+0.160**
80–90	0.239	0.440	0.469	0.489	**+0.200**
90–100	0.050	0.278	0.331	0.396	**+0.229**
100–110	0.006	0.264	0.288	0.331	**+0.258**
110–120	0.133	0.325	0.356	0.435	**+0.192**
120–130	0.223	0.389	0.411	0.434	**+0.166**
130–140	0.256	0.462	0.493	0.528	**+0.206**
140–150	0.274	0.453	0.487	0.520	**+0.179**
150–160	0.263	0.452	0.493	0.554	**+0.189**
160–170	0.222	0.388	0.410	0.478	**+0.166**
170–180	0.196	0.344	0.371	0.395	**+0.148**
180–190	0.173	0.334	0.350	0.379	**+0.161**
190–200	0.108	0.284	0.284	0.328	**+0.176**
All depths	0.389 $\pm$0.087	0.527 $\pm$0.099	0.549 $\pm$0.087	0.579 $\pm$0.078	**+0.138**
Shallow (<60 m)	0.786 $\pm$0.044	0.821 $\pm$0.042	0.831 $\pm$0.040	0.837 $\pm$0.033	**+0.035**
Deep (≥60 m)	0.219 $\pm$0.109	0.402 $\pm$0.124	0.429 $\pm$0.108	0.468 $\pm$0.098	**+0.182**

Does our train-test splitting strategy matter?

Are we getting data leakage between the train and test sets?

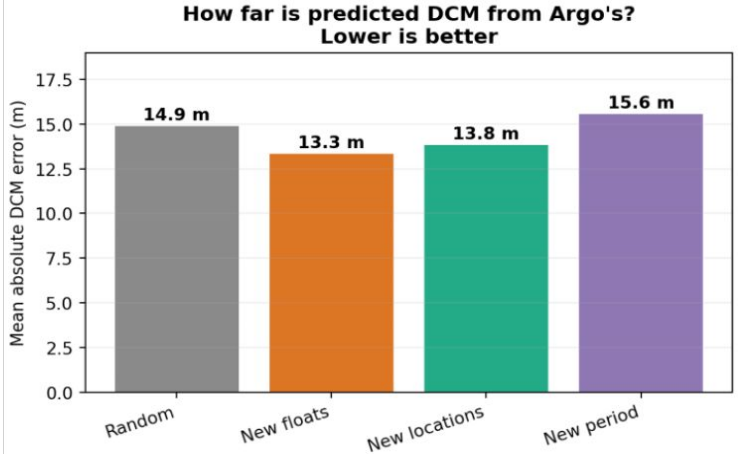
New Float, New Place, New Time—Can the Model Still Find the Depth of Maximum Chlorophyll?

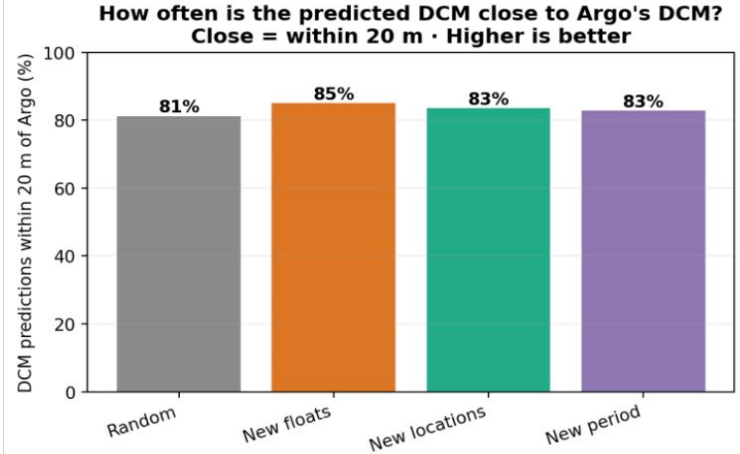
All Argo-PACE matchups



Randomly? Maybe not for Argo

Profiles from the same float are not fully independent.





BBP700-Z

Global 3D particle backscattering depth profiles from PACE, via boosted regression trees

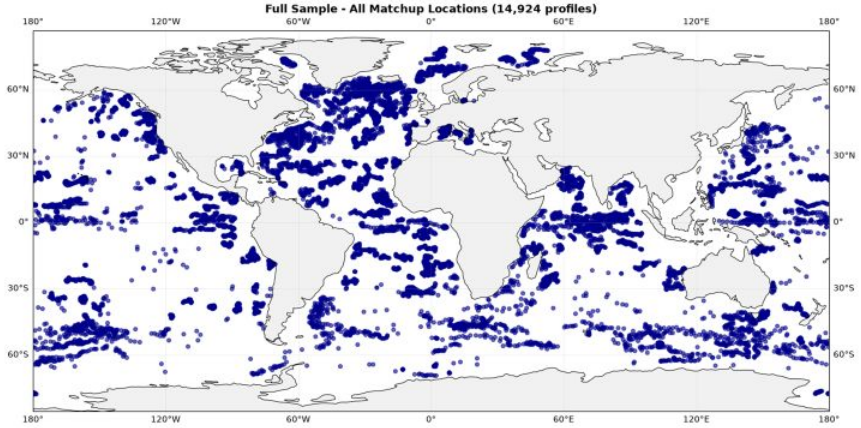
GO-BGC Hackweek 2026

Method follows Eli Holmes' CHLA-Z

Haipeng

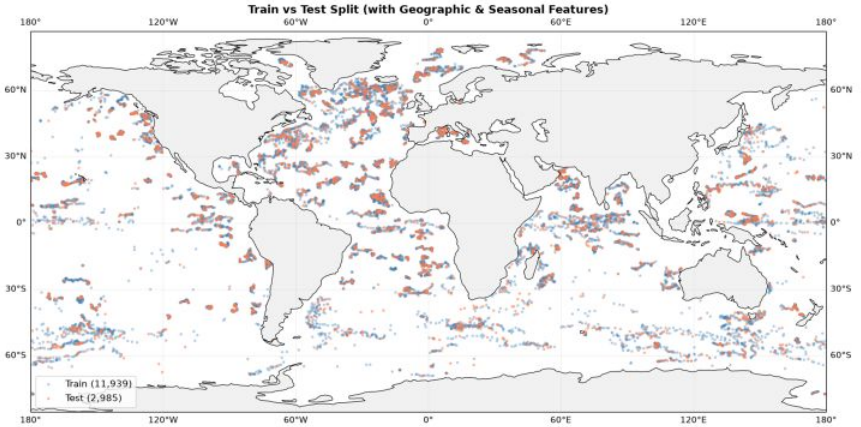
Argo × PACE Matchups

Profiles	**14,924**
PACE Rrs bands	172 (346–719 nm)
Depth bins	20 (0–200 m, 10 m intervals)
Source	BGC-Argo backscatter profiles matched to PACE OCI reflectance



Train / Test Split

Train	**11,939 profiles (80%)**
Test	2,985 profiles (20%)
Split	Random, seed=42; per-depth N varies slightly after dropping non-positive BBP700 values



Training the BRT Models

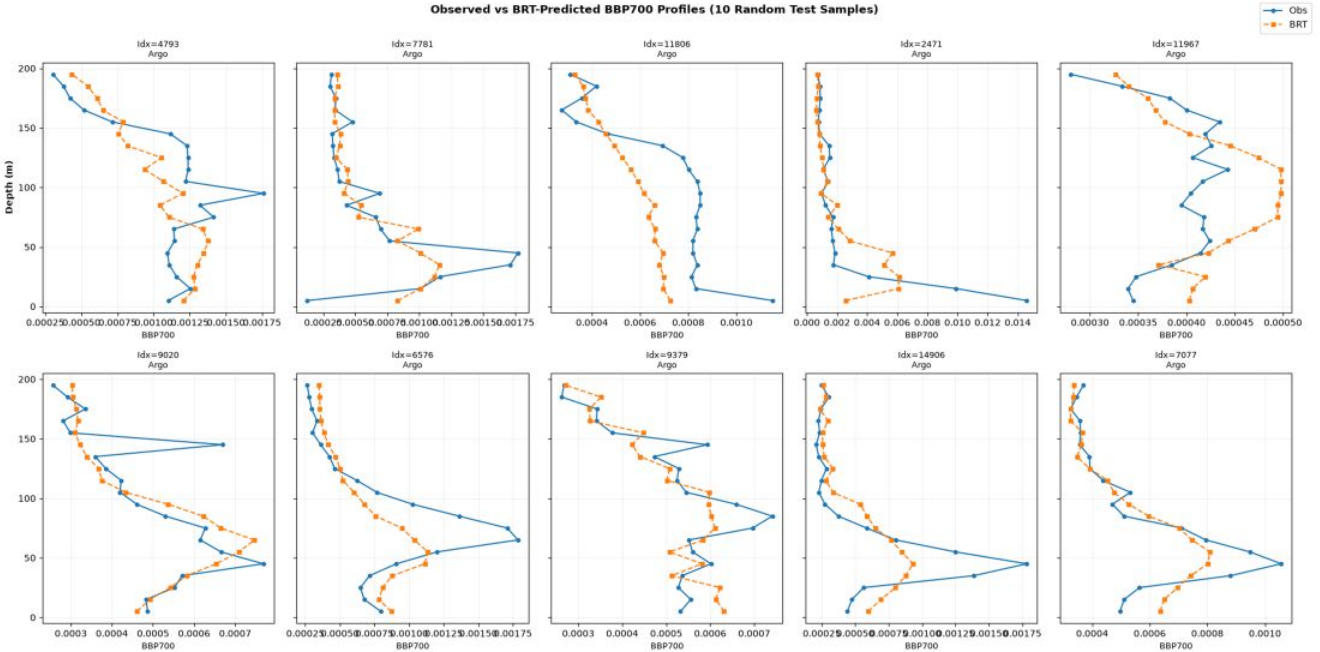
Model	**HistGradientBoostingRegressor (scikit-learn)**
Models fit	20 — one per depth bin
Target	log10(BBP700)
Features	172 Rrs bands + solar_hour, x_geo, y_geo, z_geo, day_of_year, month, type (179 total)
Hyperparameters	max_depth=6, learning_rate=0.05, max_iter=400, early_stopping=True, validation_fraction=0.1
Fit on	Training rows only (11,939)

Validation — Performance by Depth, All 20 Bins

Depth (m)	Train R^2	Train RMSE	Train N	Test R^2	Test RMSE	Test N
5	0.823	0.144	11,686	0.748	0.159	2,906
15	0.833	0.142	11,681	0.734	0.164	2,908
25	0.810	0.151	11,688	0.727	0.164	2,907
35	0.808	0.146	11,692	0.719	0.164	2,906
45	0.803	0.142	11,685	0.700	0.164	2,905
55	0.748	0.144	11,586	0.621	0.168	2,886
65	0.720	0.148	11,569	0.554	0.178	2,889
75	0.694	0.153	11,571	0.509	0.179	2,884
85	0.714	0.145	11,566	0.506	0.183	2,883
95	0.563	0.179	11,549	0.437	0.182	2,880
105	0.717	0.142	11,488	0.518	0.176	2,870
115	0.718	0.143	11,429	0.549	0.173	2,860
125	0.658	0.158	11,428	0.479	0.186	2,857
135	0.696	0.150	11,420	0.530	0.177	2,854
145	0.676	0.156	11,512	0.553	0.172	2,866
155	0.719	0.146	11,460	0.536	0.176	2,862
165	0.649	0.163	11,406	0.507	0.179	2,853
175	0.721	0.148	11,405	0.565	0.170	2,852
185	0.717	0.149	11,405	0.593	0.168	2,851
195	0.725	0.146	11,469	0.553	0.174	2,867
Mean	0.726	0.150	—	0.582	0.173	—

R^2 and RMSE on log10(BBP700); HistGradientBoostingRegressor, one model per depth bin, seed=42

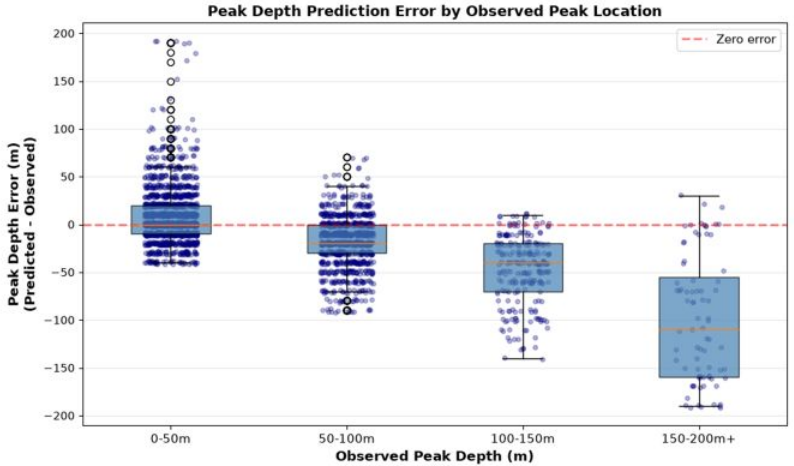
Validation — Model Fit



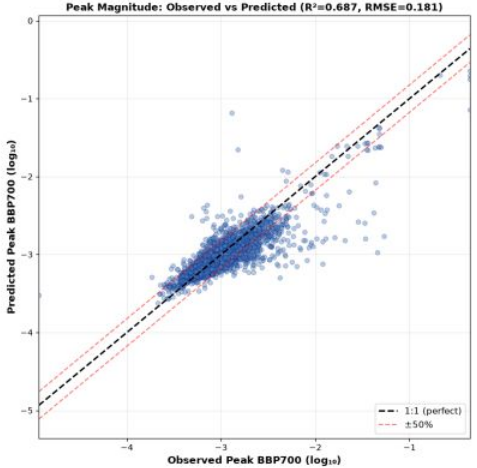
10 random test profiles, observed vs. predicted, 0–200 m

Validation — Peak Depth & Magnitude

Statistic	Peak depth, obs (m)	Peak depth, pred (m)	Peak depth error (m)	Peak val, obs	Peak val, pred	Peak val error
Count	2,923	2,923	2,923	2,923	2,923	2,923
Mean	49.749	41.798	-7.951	0.002	0.002	-0.001
Std	40.339	32.687	38.290	0.017	0.008	0.011
Median	45.000	35.000	0.000	0.001	0.001	-0.000
Min / Max	5 / 195	5 / 195	-190 / 190	-0.000 / 0.438	0.000 / 0.229	-0.367 / 0.065

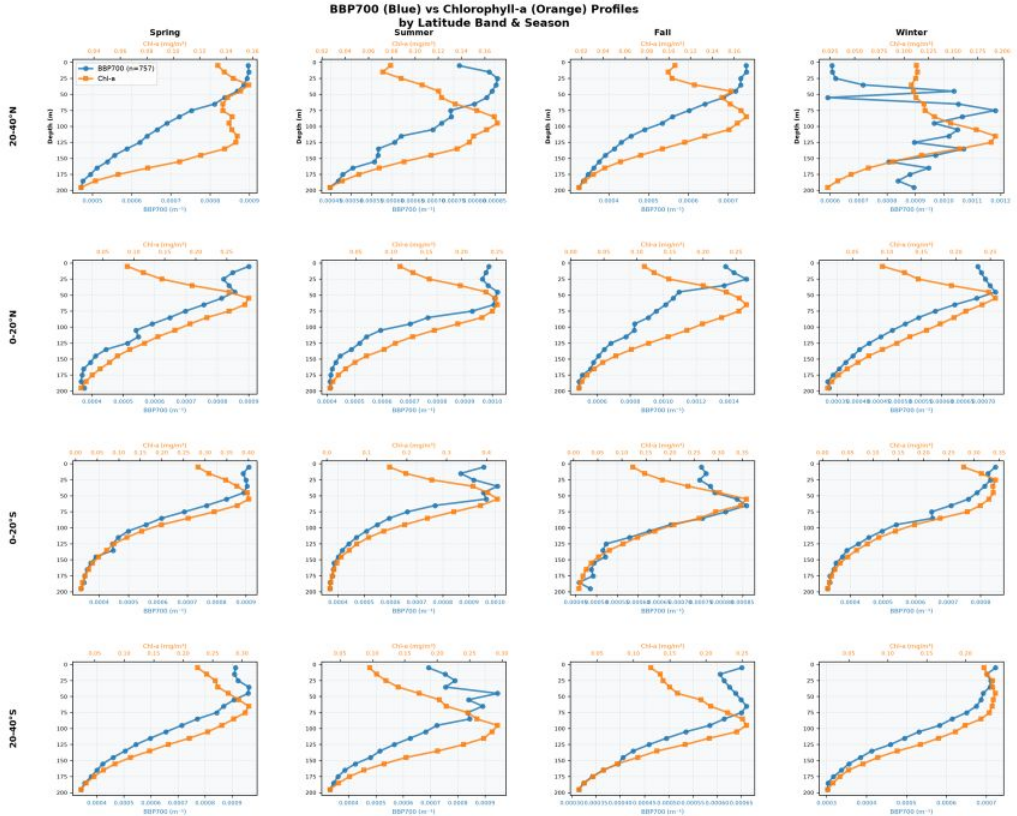


Peak depth error — within ±10 m: 47.2% · within ±20 m: 64.2%



Peak magnitude — R^2 = 0.687 (log scale)

BBP700 vs. Chlorophyll-a



BBP700 (blue) vs. chlorophyll (orange)



NEXT STEPS